TECHNICAL TRAINING CATALOGUE 2026

RENAULT TRUCKS

OUR COMMITMENT:

TO ASSIST YOU IN DEVELOPING THE COMPETENCE AND PERFORMANCE OF YOUR EMPLOYEES AND YOUR SERVICE.

Version of October 27, 2025

EDITO

Dear Subsidiary,

The Group Academy is pleased to present its catalogue of services for 2026 and to inform you of our commitments for all subsidiaries:

- ✓ **To support you in developing the skills and performance of your employees and your service,**
- ✓ **To support you in resolving technical problems on customer vehicles,**
- ✓ **To standardize and normalize the skill levels of the group's technicians.**

The diversity of brands deployed by CFAO is a real asset for the development of our skills development offers to subsidiaries throughout the company's territories. This diversity challenges us to ensure a high level of service quality within the after-sales service, the parts service, the sales department and the financial and legal service.

To improve the quality of your employees' services, we recommend deploying the following technical skills acquisition cycle:

- ✓ **Measure** the current skills of technicians and teams,
- ✓ **Exploit** the results to guide the skills development strategy,
- ✓ **Define** individual and collective skills development plans,
- ✓ **Acquire** new skills through face-to-face or remote trainings,
- ✓ **Certify** the new skills acquired,
- ✓ **Apply** new skills in a professional environment,
- ✓ **Perform** in a professional environment.

We also offer to **assist** you in **resolving technical problems** that you encounter in your workshops, with **technical support integrated** into the Academy.

The development of new skills is carried out by certified experts each in their areas of expertise, using innovative training methods.

Thank you for your commitment to developing the skills of your employees. We look forward to hearing from you.

Laurent GONIN

TEAM EQUIPMENT

The Douala Academy for EQUIPMNET is dedicated to the **RENAULT TRUCKS, JCB, LOVOL, TOYOTA FORKLIFT, BT, FUSO, HINO, SINOTRUK, SHACMAN and KING LONG** brands and is equipped with:

- A reception and a restaurant,
- Two training rooms (board, video projector, others),
- A 200 m² workshop,
- Didactic equipment for engines, transmissions and others,
- A RENAULT TRUCKS K Range truck

Our training courses are delivered at the Douala Academy and at all our subsidiaries. We also provide training at our customers' sites. Training sessions are made up of a maximum of twelve trainees.

Training service Contact:

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Thierry KOUGHA

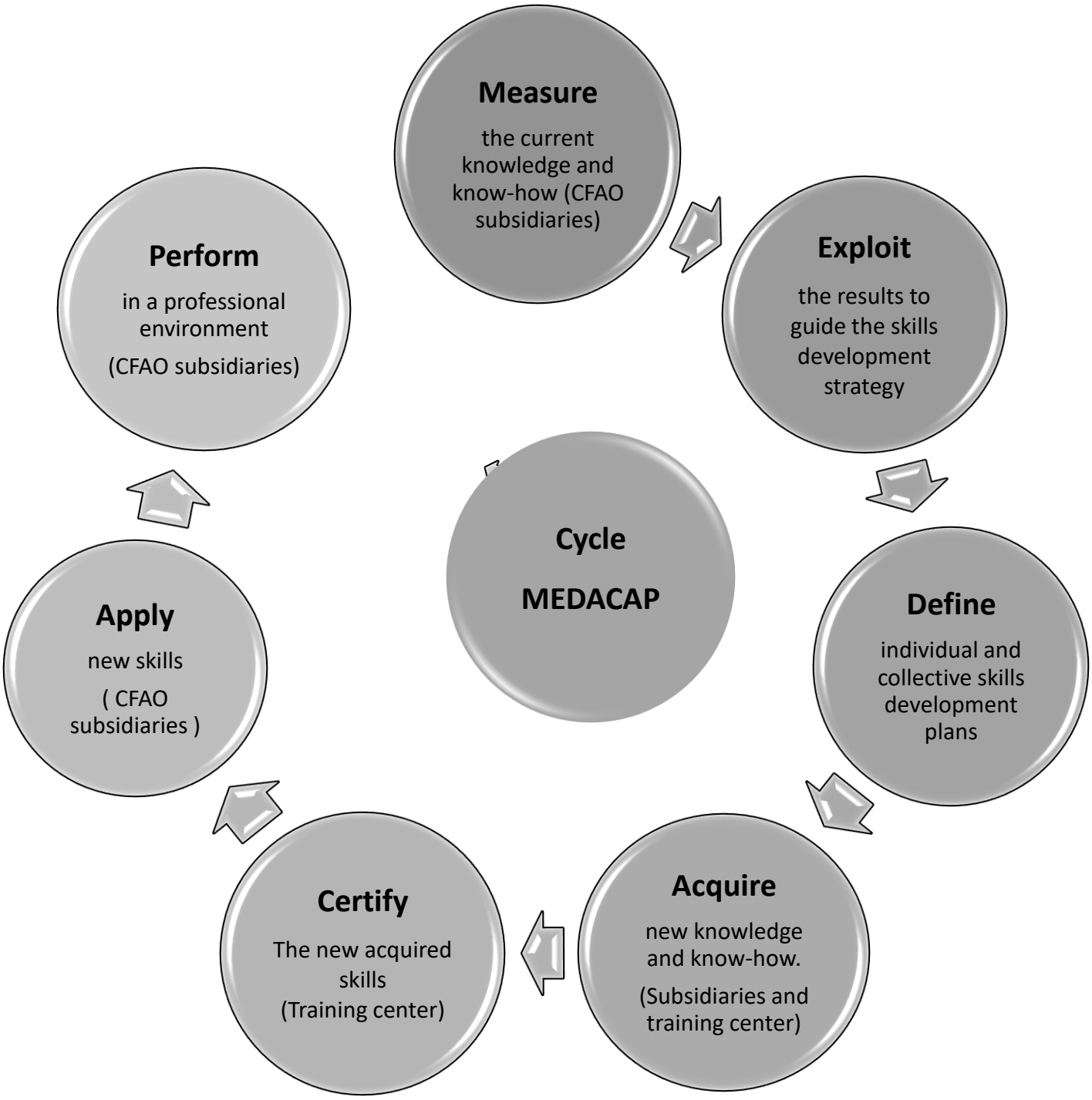
Technical Trainer Expert (TOYOTA FORKLIFT, JCB, LOVOL)



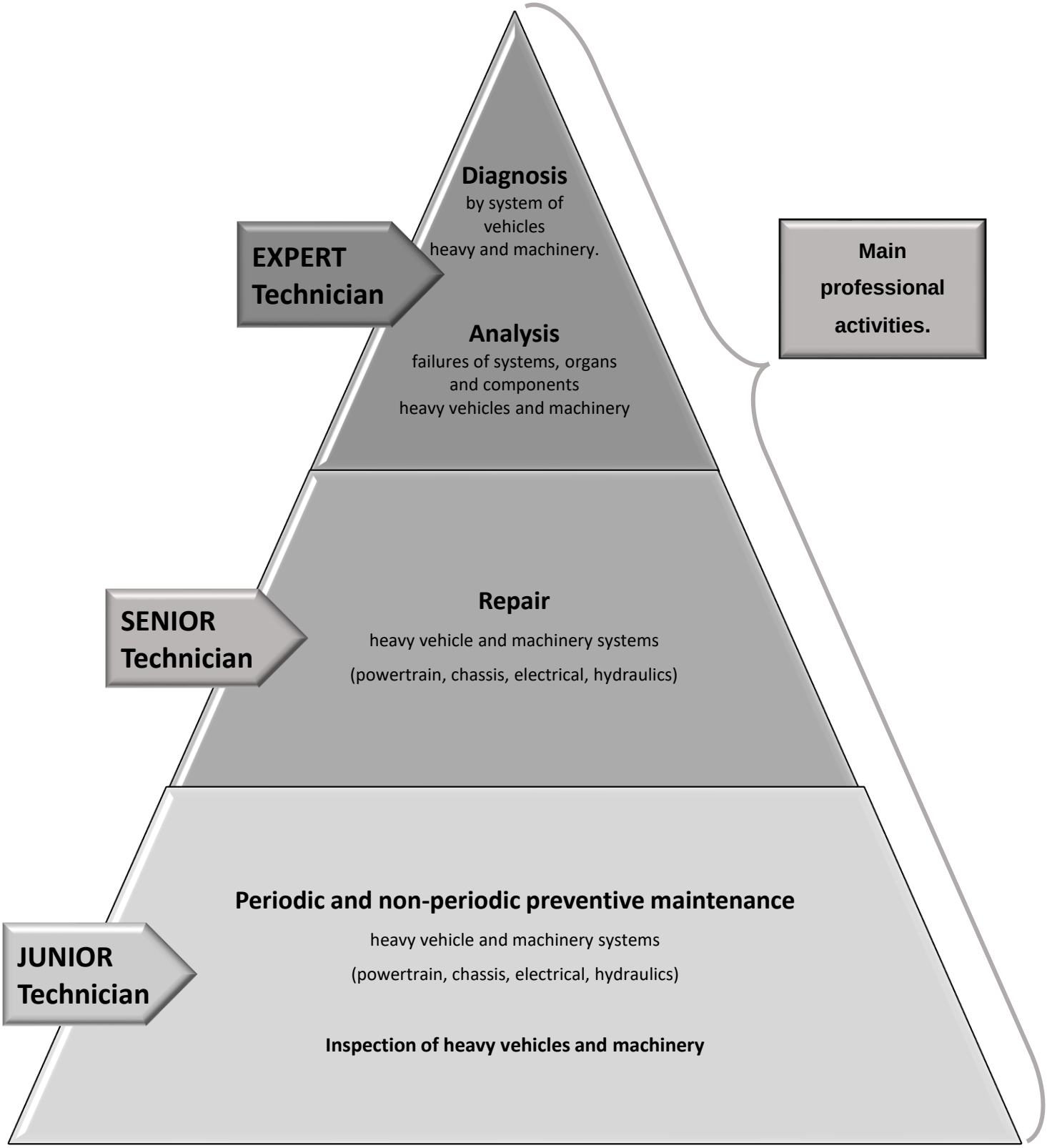
Roland NENDA

Operator Trainer
All- brands expert (Trucks and Heavy Equipment)

DEVELOPING TECHNICIANS' PROFESSIONAL EXPERTISE



SKILL LEVELS OF TECHNICIANS



TRAINING BY TECHNICIAN LEVEL

	Theoretical Training (Knowledge -> E-Learning/Distance Learning)		Practical training (Know-how -> Face-to-face)	
	Themes	**Duration (days)**	**Themes**	**Duration (days)**
EXPERT Technician	❖ The 7 stages of diagnosis applied to the different systems of the heavy vehicle (powertrain, chassis, electricity), ❖ Diag Tool Features,	20	❖ Diagnostic of DXI 11 Engine, ❖ Diagnostic of DXI 13 Engine, ❖ Diagnostic of DXI 8 Engine, ❖ Diagnostic of Optidriver Gearbox, ❖ Diagnostic of B14 Gearbox, ❖ Diagnostic of TEA2+ T2 Electrical System, ❖ DiagnosisConventional braking system with EBS,	5 5 5 5 5 5 5
SENIOR Technician	❖ Technological knowledge of the various heavy vehicle systems (powertrain, chassis, electricity). ❖ Removal and installation procedure in accordance with the manufacturer, ❖ Manufacturer-compliant disassembly and reassembly procedure ❖ Manufacturer-compliant component inspection procedure	20	❖ Repair of DXI 11 Engine, ❖ Repair of DXI 13 Engine, ❖ Repair of DXI 8 Engine, ❖ Repair of Optidriver Gearbox, ❖ Repair of Gearbox B14, ❖ Repair of Electrical System, ❖ Repair Conventional braking system with EBS,	5 5 5 5 5 5 5
JUNIOR Technician	❖ Brand history (heavy vehicle ranges, characteristics, etc.), ❖ Description of the structure of the heavy vehicle and the roles of its different systems (powertrain, chassis, electricity). ❖ Role of liquids and lubricants. ❖ Principle of use of metrology tools, ❖ Periodic and non-periodic Repair ofheavy vehicles, ❖ The fundamental principles of engine, transmission, electricity and hydraulics, ❖ Others.	25	❖ Repair ofK range heavy vehicle systems, ❖ Repair ofDXI 11 engine, ❖ Repair ofDXI 13 engine, ❖ Repair ofDXI 8 engine, ❖ Repair ofOptidriver gearbox, ❖ Repair ofB14 Gearbox,	5 3 3 3 3 3

Page	Code	Training designation	Date	Place
8 to 14	Repair of heavy vehicle systems			
9	RT01	Repair of K-series truck systems	To be defined	To be defined
10	RT02	Repair ofthe DXI11 engine	To be defined	To be defined
11	RT03	Repair ofthe DXI13 engine	To be defined	To be defined
12	RT04	Repair ofthe DXI8 engine	To be defined	To be defined
13	RT05	Optidriver gearbox	To be defined	To be defined
14	RT06	Repair ofthe B14 gearbox	To be defined	To be defined
15 to 22	Repair / Overhaul of heavy vehicle systems			
16	RT07	DXI 11 Engine Repair/Overhaul	To be defined	To be defined
17	RT08	DXI 13 Engine Repair/Overhaul	To be defined	To be defined
18	RT09	DXI 18 engine repair/overhaul	To be defined	To be defined
19	RT10	Optidriver gearbox repair/overhaul	To be defined	To be defined
20	RT11	B14 gearbox repair/overhaul	To be defined	To be defined
21	RT12	Repair of vehicle electrical systems	To be defined	To be defined
22	RT13	Repair/Overhaul of the air brake system with EBS	To be defined	To be defined
23	RT14	Corrective maintenance of components for heavy vehicles in the K range	To be defined	To be defined
24 to 32	Heavy vehicle systems diagnosis			
25	RT15	DXI11 engine diagnosis	To be defined	To be defined
26	RT16	DXI13 engine diagnosis	To be defined	To be defined
27	RT17	DXI8 engine diagnosis	To be defined	To be defined
28	RT18	Optidriver gearbox diagnosis	To be defined	To be defined
29	RT19	Diagnosis of B14 gearbox	To be defined	To be defined
30	RT20	Diagnosis of vehicle electrical and electronic systems	To be defined	To be defined
31	RT21	Brake system diagnosiswith EBS	To be defined	To be defined
32	RT22	Electrical System DiagnosisTEA2+ T2	To be defined	5 days

REPAIR OF HEAVY VEHICLE SYSTEMS

(JUNIOR Technician)



RT01	REPAIR OF K-RANGE TRUCK SYSTEMS

Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** systems (powertrain, chassis, electrical) in operation according to the manufacturer's recommendations, • **Inspect** the systems (powertrain, chassis, electrical) according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Truck ranges and their characteristics • Description of the vehicle structure and role of the different truck systems (Powertrain, Chassis, Electricity). • Roles of liquids and lubricants • Principles of using metrology tools • The different types of maintenance • Repair ofsystems ➢ Periodicals, ➢ Non-periodicals ► **Part Practical:** • Replace oils and fluids (engine, gearbox, axles, other) • Top up oils and fluids (engine, gearbox, axles, other) • Remove and refit components (belt and accessories, brake pads or linings, etc.) according to the manufacturer's recommendations. • Check the conformity of the components (dimensions, flatness, voltage, current) to the manufacturer's expectations before reassembling them. • Visually inspect the truck systems ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and Place	To be defined with the client	To be defined with the client	

RT02	REPAIR OF THE DXI 11L ENGINE

Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** the DXI 11L engine of the K range in operation according to the manufacturer's recommendations. • **Inspect** the DXI 11L engine of the K range according to the manufacturer's recommendations.		
Description Theory (30%) Practical (70%)	► **Part Theoretical:** • Description and operation of the DXI 11L engine, • Description of lubrication and cooling systems, • Role of liquids and lubricants, • Description of the injection system, • The different types of maintenance, • Repair ofthe DXI 11L engine from the K range, ➢ Periodicals, ➢ Non-periodicals, ► **Part Practical:** • Replace or top up the engine oil, • Remove and reinstall components (filters, etc.) according to the manufacturer's recommendations. • Check that the components (engine silent block , engine mounting bolts, radiator mounting, hoses, oil pan) conform to the manufacturer's specifications . • Visually inspect the engine. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT03	REPAIR OF THE DXI 13L ENGINE

Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** the DXI 13L engine of the K range in operation according to the manufacturer's recommendations. • **Inspect** the DXI 13L engine of the K range according to the manufacturer's recommendations.		
Description Theory (30%) Practical (70%)	► **Part Theoretical:** • Description and operation of the DXI 13L engine, • Description of lubrication and cooling systems, • Role of liquids and lubricants, • Description of the injection system, • The different types of maintenance, • Repair ofthe DXI 13L engine from the K range, ➢ Periodicals, ➢ Non-periodicals, ► **Part Practical:** • Replace or top up the engine oil, • Remove and reinstall components (filters, etc.) according to the manufacturer's recommendations. • Check that the components (engine mounts , engine mounting bolts, radiator mounting, hoses, oil pan, etc.) conform to the manufacturer's specifications . • Visually inspect the engine. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT04	REPAIR OF THE DXI 8L ENGINE

Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** the DXI 8L engine of the D range in operation according to the manufacturer's recommendations. • **Inspect** the DXI 8L engine of the D range according to the manufacturer's recommendations.		
Description Theory (30%) Practical (70%)	► **Part Theoretical:** • Description and operation of the DXI 8L engine, • Description of lubrication and cooling systems, • Role of liquids and lubricants, • Description of the injection system, • The different types of maintenance, • Repair ofthe DXI 8L engine from the D range, ➢ Periodicals, ➢ Non-periodicals, ► **Part Practical:** • Replace or top up the engine oil, • Remove and reinstall components (filters, etc.) according to the manufacturer's recommendations. • Check the conformity of the components (engine silent block , engine fixing screws, radiator fixing, hoses, oil pan…) to the manufacturer 's expectations • Visually inspect the engine. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT05	RRPAIR OF THE OPTIDRIVER GEARBOX

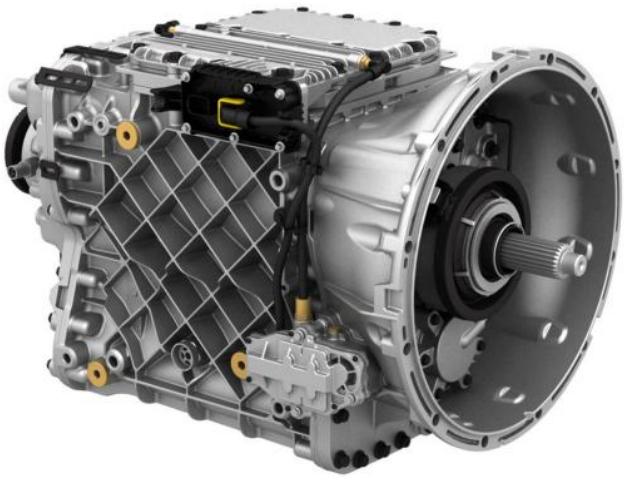
Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** OPTIDRIVER K-series gearboxes in operation according to the manufacturer's recommendations. • **Inspect** OPTIDRIVER gearboxes of the K range according to the manufacturer's recommendations.		
Description Theory (30%) Practical (70%)	► **Part Theoretical:** • Types of gearboxes in the K range and their characteristics, • Description and operation of the OPTIDRIVER gearbox , • Role of lubricants, • The different types of maintenance • Repair ofOPTIDRIVER gearboxes (K range) , ➢ Periodicals, ➢ Non-periodicals, ► **Part Practical:** • Replacing or topping up the Optidriver gearbox oil , • Remove and reinstall components (filters, etc.) according to the manufacturer's recommendations. • Check that the components conform to the manufacturer's specifications before reassembling them. • Visually inspect the gearbox . ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT06	REPAIR OF THE B14 GEARBOX

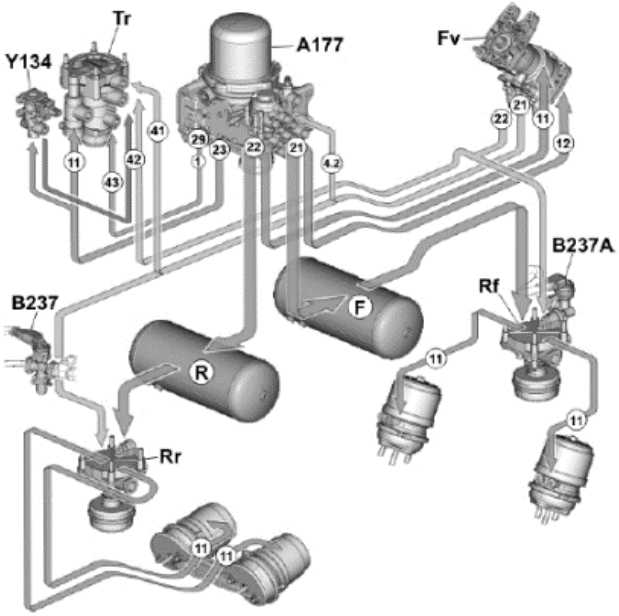
Level, Mode and Duration	JUNIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Maintain** the B14 K series gearboxes in operation according to the manufacturer's recommendations. • **Inspect** the B14 gearboxes of the K range according to the manufacturer's recommendations.		
Description Theory (30%) Practical (70%)	► **Part Theoretical:** • Types of gearboxes in the K range and their characteristics, • Description and operation of the B14 gearbox , • Role of lubricants, • The different types of maintenance • Repair ofB14 gearboxes (K range) , ➢ Periodicals, ➢ Non-periodicals, ► **Part Practical:** • Replace or top up the B14 gearbox oil , • Remove and reinstall components (filters, etc.) according to the manufacturer's recommendations. • Check that the components conform to the manufacturer's specifications before reassembling them. • Visually inspect the gearbox . ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technician currently being integrated, requiring operational skills in preventive maintenance		
Prerequisites	Having completed basic mechanics training		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

REPAIR / OVERHAUL OF HEAVY VEHICLE SYSTEMS

(SENIOR Technician)







RT07	DXI 11L ENGINE REPAIR/OVERHAUL

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the DXI 11L engine, • **Disassemble** and **reassemble** the DXI 11L engine according to the manufacturer's instructions. • **Checking** for compliance DXI 11L engine components with the standard values recommended by the manufacturer, • **Test and adjust** the DXI 11L engine according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 11L engine • Description of the lubrication and cooling system, • Description of the injection system, • Description of the electronic engine management system, • Removal and installation procedure • Disassembly and reassembly procedure, • Principle of using metrology tools. ► **Part Practical:** • Remove the DXI 11L engine according to the manufacturer's instructions . • Disassemble the DXI 11L engine according to the manufacturer's instructions . • Clean the components of the DXI 11L engine, • Visual and geometric inspection of components, • Verify that the components conform to the manufacturer's specifications . • Replace components that do not meet the manufacturer's specifications . • Reassemble the engine components according to the manufacturer's instructions . • Place the engine back on the truck. • Engine testing and adjustments on the truck. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Organs Technician		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, DXI 11L engine on bench, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT08	ENGINE REPAIR / OVERHAUL OF THE 13L DXI

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the DXI 13L engine, • **Disassemble** and **reassemble** the DXI 13L engine according to the manufacturer's instructions. • **Checking** for compliance DXI 13L engine components with the standard values recommended by the manufacturer, • **Test and adjust** the DXI 13L engine according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 13L engine • Description of the lubrication and cooling system, • Description of the injection system, • Description of the electronic engine management system, • Removal and installation procedure • Disassembly and reassembly procedure, • Principle of using metrology tools. ► **Part Practical:** • Remove the DXI 13L engine according to the manufacturer's instructions . • Disassemble the DXI 13L engine according to the manufacturer's instructions . • Clean the components of the DXI 13L engine, • Visual and geometric inspection of components, • Verify that the components conform to the manufacturer's specifications . • Replace components that do not meet the manufacturer's specifications . • Reassemble the engine components according to the manufacturer's instructions . • Place the engine back on the truck. • Engine testing and adjustments on the truck. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Organs Technician		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, DXI 13L engine on bench, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT09	DXI 8L ENGINE REPAIR/OVERHAUL

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the DXI 8L engine, • **Disassemble** and **reassemble** the DXI 8L engine according to the manufacturer's instructions. • **Checking** for compliance DXI 8L engine components with the standard values recommended by the manufacturer, • **Test and adjust** the DXI 8L engine according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 8L engine • Description of the lubrication and cooling system, • Description of the injection system, • Description of the electronic engine management system, • Removal and installation procedure • Disassembly and reassembly procedure, • Principle of using metrology tools. ► **Part Practical:** • Remove the DXI 8L engine according to the manufacturer's instructions . • Disassemble the DXI 8L engine according to the manufacturer's instructions . • Clean the components of the DXI 8L engine, • Visual and geometric inspection of components, • Verify that the components conform to the manufacturer's specifications . • Replace components that do not meet the manufacturer's specifications . • Reassemble the engine components according to the manufacturer's instructions . • Place the engine back on the truck. • Engine testing and adjustments on the truck. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Organs Technician		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, DXI 8L engine on bench, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT10	OPTIDRIVER GEARBOX REPAIR/OVERHAUL

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Place** and **set aside** the box of speeds OPTIDRIVER , • **Disassemble** and **reassemble** the box of speeds OPTIDRIVER , following the manufacturer's instructions • **Checking** for compliance components of the box of speeds OPTIDRIVER with the standard values recommended by the manufacturer, • **Test and adjust** the box of speeds OPTIDRIVER while respecting the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the box of speeds OPTIDRIVER • Removal and installation procedure • Disassembly and reassembly procedure, • Principle of using metrology tools. ► **Part Practical:** • Submit the box of speeds OPTIDRIVER following the manufacturer's instructions , • Disassemble the box of speeds OPTIDRIVER following the manufacturer's instructions , • Clean the components of the box of speeds OPTIDRIVER , • Visual and geometric inspection of components, • Verify that the components conform to the manufacturer's specifications. • Replace components that do not meet the manufacturer's specifications . • Reassemble the engine components according to the manufacturer's instructions . • Rest the box of speeds OPTIDRIVER on the truck, • Gearbox testing and adjustments of speeds OPTIDRIVER on the truck. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Organs Technician		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, Box of speeds OPTIDRIVER on bench, Adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT11	B14 GEARBOX REPAIR/OVERHAUL

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Place** and **set aside** the box of speeds B14 , • **Disassemble** and **reassemble** the box of speeds B14, following the manufacturer's instructions. • **Checking** for compliance components of the box of speeds B14 with the standard values recommended by the manufacturer, • **Test and adjust** the box of speeds B14 while respecting the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the box of speeds B14, • Removal and installation procedure • Disassembly and reassembly procedure, • Principle of using metrology tools. ► **Part Practical:** • Submit the box of speeds B14 following the manufacturer's instructions , • Disassemble the box of speeds B14 following the manufacturer's instructions , • Clean the components of the box of speeds B14 , • Visual and geometric inspection of components, • Verify that the components conform to the manufacturer's specifications. • Replace components that do not meet the manufacturer's specifications . • Reassemble the engine components according to the manufacturer's instructions . • Rest the box of speeds B14 on the truck, • Gearbox testing and adjustments of speeds B14 on the truck. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Organs Technician		
Prerequisites	Have acquired the skills of the JUNIOR Technician Level (Maintenance)		
Language	French		
Equipment	Training room and workshop equipped, Box of speeds B14 on bench, Adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT12	ELECTRICAL SYSTEM REPAIR/OVERHAUL

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the starter and alternator . • **Disassemble** and **reassemble** the starter and alternator, following the manufacturer's instructions. • **Checking** for compliance starter and alternator components with the standard values recommended by the manufacturer, • **Repairing** electrical wires .		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • A review of the principles of electricity and electronics, • Description and operation of the starting system, • Description and operation of the charging system, • Procedure for removing and installing the starter and alternator, • Starter and alternator disassembly procedure, • Electrical wire repair procedure • Principle of use of measuring tools. ► **Part Practical:** • Remove and reinstall the starter and alternator according to the manufacturer's instructions. • Remove and reinstall the electrical wiring harnesses according to the manufacturer's instructions. • Disassemble and reassemble the starter and alternator, • Clean the starter and alternator components, • Visual and geometric inspection of components (starter / alternator), • Verify that the components conform to the manufacturer's specifications. • Replace components that do not meet the manufacturer's specifications. • Reinstall the starter/alternator, • Test the operation of the starter/alternator. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Electrical Systems Technician		
Prerequisites	Having completed basic training in mechanics, electricity and electronics		
Language	French		
Equipment	Equipped training room and workshop, starter, alternator, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT13	REPAIR / OVERHAUL OF THE AIR BRAKE SYSTEM (WITH EBS)

Level, Mode and Duration	SENIOR	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the drum brake mechanism, • **Disassemble** and **reassemble** drum brake mechanisms according to the manufacturer's recommendations. • **Check** the drum brake components against the standard values recommended by the manufacturer. • **Submit** and **revise** the APM2, • **Test** and **adjust** the Z cam brake mechanism,		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the pneumatic braking system, • Types of brakes, • Cam brake systems, • Description and operation of the Z cam brake, • Procedure for removing and installing the Z cam brake, • Procedure for disassembling and reassembling the Z cam brake, ► **Part Practical:** • Remove and reinstall a relay valve, following the manufacturer's recommendations. • Remove and reinstall the APM2 according to the manufacturer's recommendations. • Disassemble and reassemble the parking brake relay valve incorporated in the APM2, • Remove the Z cam brake mechanism, • Disassemble the Z cam brake mechanism, • Clean the components of the Z cam brake mechanism, • Visual and geometric inspection of components (brake drum, etc.), • Verify that the components conform to the manufacturer's specifications. • Replace components that do not meet the manufacturer's specifications. • Reassemble the components of the Z cam brake mechanism, • Reinstall the Z cam brake mechanism, • Test and adjust the Z cam brake. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Qualified technicians with skills in mechanics, electricity, pneumatics		
Prerequisites	basic mechanics and pneumatics training		
Language	French		
Equipment	Equipped training room and workshop, starter, alternator, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT14	CORRECTIVE MAINTENANCE OF COMPONENTS OF K-RANGE HEAVY VEHICLES (SDR)

Level, Mode and Duration	SENIOR	Face-to-face	04 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Overhaul and/or repair of components of heavy vehicles in the K range (DXI 11L/13L engine,** Optidriver gearbox , EBS air braking, electricity and air conditioning) , • Training required for **SDR certification.**		
Description Theory (20%) Practical (80%)	► **Part Theoretical:** • Optidriver gearbox , EBS air braking, electricity and air conditioning) , **Inspection and repair procedure** according to the manufacturer's recommendations : - Remove and install the above-mentioned components. - Disassembly and reassembly - Component control - Testing of organs under operating conditions ► **Part Practical:** • Overhaul and/or repair of the DXI 11L/13L engine from the K range according to the manufacturer's recommendations. • Optidriver gearbox from the K range according to the manufacturer's recommendations. • Overhaul and/or repair of the EBS pneumatic braking system of the K range according to the manufacturer's recommendations. • Overhaul and/or repair of the K range electrical system according to the manufacturer's recommendations. • Overhaul and/or repair of the K series air conditioning system according to the manufacturer's recommendations. ► **Evaluation part:** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Renault Trucks online test (712610), training modules from RT07 to RT13		
Language	French		
Equipment	Training room and workshop equipped, vehicle(s) and suitable bench components available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	**To be defined**	**To be defined**	

HEAVY VEHICLE SYSTEMS DIAGNOSIS

(EXPERT Technician)



RT15	DXI 11L ENGINE DIAGNOSIS

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the DXI 11L engine, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 11L engine, • Description of the injection system, • Description of the electronic engine management system, • Malfunctions of the components of the DXI 11L engine, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Test or verify the conformity of the systems and components of the DXI 11L engine. 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired , 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (Systems Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, DXI 11L engine on bench, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT16	13L DXI ENGINE DIAGNOSIS

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the DXI 13L engine, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 13L engine, • Description of the injection system, • Description of the electronic engine management system, • Malfunctions of the components of the DXI 13L engine, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Test or verify the conformity of the systems and components of the DXI 13L engine. 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (Systems Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, DXI 13L engine on bench, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT17	DIAGNOSIS OF THE DXI 8L ENGINE

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** faults related to the DXI 8L engine, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the DXI 8L engine, • Description of the injection system, • Description of the electronic engine management system, • Malfunctions of the components of the DXI 8L engine, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Test or verify the conformity of the systems and components of the DXI 8L engine. 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (Systems Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, DXI 8L engine on bench, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT18	OPTIDRIVER GEARBOX DIAGNOSIS

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the OPTIDRIVER gearbox, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Review of the principles of electricity and k-band multiplexing, • Features and operation of the OPTIDRIVER gearbox, • Malfunctions of the OPTIDRIVER gearbox components, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Test or check the conformity of the OPTIDRIVER gearbox systems and components, 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (System Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, OPTIDRIVER gearbox on bench, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT19	DIAGNOSIS OF B14 GEARBOX

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** faults related to the B14 gearbox, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Review of the principles of electricity and k-band multiplexing, • Features and operation of the B14 gearbox, • Malfunctions of the B14 gearbox components, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Test or check the conformity of the systems and components of the B14 gearbox, 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (System Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, B14 gearbox on bench, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT20	DIAGNOSIS OF VEHICLE ELECTRICAL AND ELECTRONIC SYSTEMS

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** failures related to the TEA2+ electrical and electronic systems, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • A reminder of the principles of electricity and multiplexing, • Description and operation of the TEA2+ electronic architecture • Description and functions of the modules of the TEA2+ electronic architecture, • Malfunctions of the TEA2+ electronic architecture modules, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. Testing or verifying the conformity of the systems and components of the TEA2+ electronic architecture, 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (System Repair/Overhaul)		
Language	French		
Equipment	Equipped training room and workshop, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT21	DIAGNOSIS OF THE AIR BRAKE SYSTEM (EBS)

Level, Mode and Duration	EXPERT	In person	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the pneumatic and electronic braking system, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Review of physical principles (Flow rate, pressure) • Characteristics and operation of the components of the pneumatic and electronic braking system, • Malfunctions of the components of the pneumatic and electronic braking system, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. To test or verify the conformity of the systems and components of the pneumatic and electronic braking system, 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Theory (prerequisites and knowledge) • Practical skills (prerequisites and know-how)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a Senior Technician (Repair/overhaul of systems)		
Language	French		
Equipment	Equipped training room and workshop, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined with the client	

RT22	ELECTRICAL SYSTEM DIAGNOSISTEA2+ T2

Level, Mode and Duration	SENIOR	Face-to-face	05 days
Presenter	**RENAULT TRUCKS Expert Technical Trainer**		
Goals	• **Diagnosing** faults related to TEA2+ T2 electrical systems, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical:** • Description and operation of the TEA2+ T2 electrical architecture • Description and functions of the modules of the TEA2+ T2 electrical architecture, • Malfunctions of the modules of the TEA2+ T2 electronic architecture, • The diagnostic steps and features of the Diag tool. ► **Part Practical:** 8. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 9. Formulate hypotheses regarding the cause of the malfunction/failure, 10. Test or verify the conformity of the systems and components of the TEA2+ T2 electrical architecture with the manufacturer's recommendations. 11. Identify the most likely hypothesis based on the tests and symptoms, 12. Develop a recommendation for the components to be replaced or repaired, 13. Perform the repair of the defective component, 14. Test and adjust the system involved in the truck repair. ► **Evaluation part:** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a SENIOR Technician (Repair)		
Language	French		
Equipment	Equipped training room and workshop, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	**To be defined**	**To be defined**	